50Days-coding-challenge

Day-28[Python]

Student Marks Analyzer (Dictionary)

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Problem Statement

You are given student marks in different subjects. Use a **dictionary** to:

1. Store student data
2. Calculate:
   - Total marks
   - Average marks
3. Find:
   - Top scorer
   - Failed students (marks < 40 in any subject)

Input Example

students = {

"Asha": {"Math": 78, "Science": 85, "English": 67},

"Ravi": {"Math": 35, "Science": 40, "English": 50}, "John": {"Math": 90, "Science": 92, "English": 88}

}

Tasks

1. **Calculate total & average for each student**

```python
# 1. Calculate total & average
for student, marks in students.items():
    total = sum(marks.values())
    average = total / len(marks)

    totals[student] = total
    averages[student] = average

    print(f"{student} -> Total: {total}, Average: {average:.2f}")
    for subject, mark in marks.items():
        if mark < 40:
            failed_students.append(student)
            break  # No need to check further subjects
```

```
C:\Users\NITHIN\PycharmProjects\helloworld\.venv\Scripts\python.exe C:\Users\NITHIN\PycharmProjects\helloworld\Day29.py
Asha -> Total: 230, Average: 76.67
Ravi -> Total: 125, Average: 41.67
John -> Total: 270, Average: 90.00

Process finished with exit code 0
```

2. Identify top scorer

```python
    # 3. Check for failed students
    for subject, mark in marks.items():
        if mark < 40:
            failed_students.append(student)
            break  # No need to check further subjects

# 2. Identify top scorer
top_scorer = max(totals, key=totals.get)

print("\nTop Scorer:", top_scorer, "with", totals[top_scorer], "marks")
```

```
Ravi -> Total: 125, Average: 41.67
John -> Total: 270, Average: 90.00

Top Scorer: John with 270 marks

Process finished with exit code 0
```

3. Find students who failed in any subject

Day26.py | C:\...\Day26.py | Day27.py | C:\...\Day27.py | Day28.py | C:\...\Day28.py | Pythonass1.py | Day29.py

```python
    # 3. Check for failed students
    for subject, mark in marks.items():
        if mark < 40:
            failed_students.append(student)
            break  # No need to check further subjects

# 2. Identify top scorer
top_scorer = max(totals, key=totals.get)

print("\nTop Scorer:", top_scorer, "with", totals[top_scorer], "marks")

# 3. Failed students
print("Failed Students:", failed_students)
```

Run | Day29

```
John -> Total: 270, Average: 90.00

Top Scorer: John with 270 marks
Failed Students: ['Ravi']

Process finished with exit code 0
```

helloworld > Day29.py 28:41 CRLF UTF-8 4 spaces Python 3.13 (helloworld)